FedMCA: Cross-Attention–Based Personalized Multimodal Federated Learning under Modality Heterogeneity

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Abstract

Multimodal federated learning (MFL) often suffers from modality heterogeneity and missing modalities across clients, which can easily cause representation misalignment and unstable aggregation. We study multimodal FL under modality heterogeneity and present FedMCA, a personalized framework built on cross-attention. FedMCA improves global multimodal fusion and preserves client-level personalization without sharing raw data. The design follows a decoupled server–client paradigm. Clients maintain modality-specific encoders together with personalized decoders, whereas the server learns a fusion decoder using a publicly available full-modality dataset. We further introduce modality-level contrastive learning on the client side. This mechanism aligns local and global representations and improves robustness when modalities are missing.

Keywords: Multimodal Federated Learning; Cross-Attention; Personalization; Modality Heterogeneity; Contrastive Learning

1 Introduction

Federated learning (FL)[1] enables collaborative model training without sharing raw data, making it suitable for privacy-sensitive applications. However, most existing FL methods assume homogeneous data distributions across clients, where clients are expected to learn similar representations and optimize a shared model under comparable data settings. In real-world deployments, however, clients are often equipped with heterogeneous sensors and collect different types of data, leading to diverse data distributions and modality availability across participants.

For example, in intelligent healthcare applications, one client may possess MRI scans, while another primarily holds CT images. Such discrepancies in modality distribution introduce modality mismatch during training. Moreover, different modalities vary in representation formats, feature spaces, and semantic characteristics. Simply aggregating client models without accounting for these differences may result in information loss or even negative transfer.

Modality heterogeneity and data imbalance pose new challenges to federated learning, including inconsistent optimization and biased learning, as well as increased computational and communication costs. Most existing *multimodal* federated learning (MFL) systems adopt conventional FL aggregation with identical model architectures [2], which tends to overlook personalization and limits performance under heterogeneous settings.

In this paper, we propose a novel personalized MFL framework, as illustrated in Figure 1, to resolve the above problems. The framework is designed to handle modality heterogeneity and missing modalities while keeping client data private. Specifically, the server performs cross-attention–based multimodal fusion by aggregating encoder representations and learning a global multimodal decoder using a publicly available full-modality dataset. On the client side, each participant maintains a personalized encoder–decoder model for local feature extraction and prediction, enabling adaptation to heterogeneous modalities while preserving data privacy.

Our main contributions are:

- We propose a cross-attention mechanism for server-side multimodal fusion and introduce task-aware guidance from downstream models into the fusion process. By coupling attention with task-level supervision, we promote semantically meaningful alignment across modalities and achieve improved robustness in heterogeneous and missing-modality scenarios.
- We design a global–local multimodal contrastive learning mechanism to strengthen semantic alignment, mitigate representation gaps in modality-specific encoders under missing-modality settings, and improve the effectiveness of subsequent multimodal fusion.
- We facilitate a personalized MFL scheme that jointly improves multimodal fusion performance and preserves client privacy while enabling clients to perform personalized optimization according to their local data characteristics without sharing sensitive information.

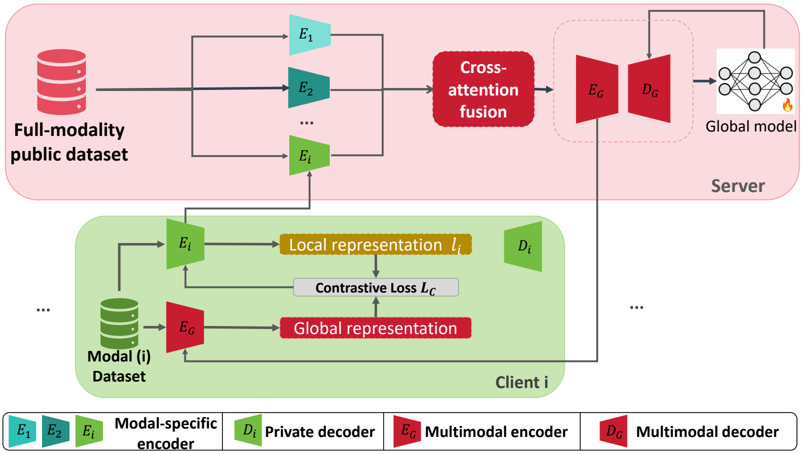

Fig. 1 Overall framework of personalized MFL

2 Related Work

Unlike traditional FL, MFL considers clients with multimodal data, but modality heterogeneity remains challenging to date [3]. It mainly involves a *modality gap* (unimodal clients cannot align with unseen modalities) and a *task gap* (clients optimize different objectives, e.g., classification vs. retrieval), which exacerbates aggregation drift and degrades the global model.

To mitigate modality heterogeneity, various methods have been proposed. MMFed [4] develops a unified MFL architecture and introduces a co-attention mechanism to dynamically fuse multimodal features, improving global performance on multimodal tasks. Nevertheless, MMFed requires uploading the parameters of both the attention module and the classifier to the server in each training round, which increases communication overhead and may become a bottleneck in bandwidth-limited networks.

CreamFL [5] employs knowledge distillation and contrastive aggregation to mitigate model drift, but struggles under non-IID distributions and unimodal-dominant settings.FedMEMA [6] employs modality-specific encoders and global anchors for multimodal alignment, but introduces high computational complexity and requires full-modality data at the server.

For medical multimodal settings, FedMEMA [6] proposes modality-specific encoders to alleviate cross-modal heterogeneity and introduces global anchors with local calibration modules to compensate for missing modalities. Despite its effectiveness, FedMEMA relies on multiple complex components, increasing the computational burden on both the server and clients. In addition, it assumes that the server has access to complete multimodal data for optimizing the global model, which may not hold in privacy-sensitive scenarios.

In general, it remains a challenge to design an MFL framework that is efficient, robust, personalized, and communication-efficient under modality heterogeneity.

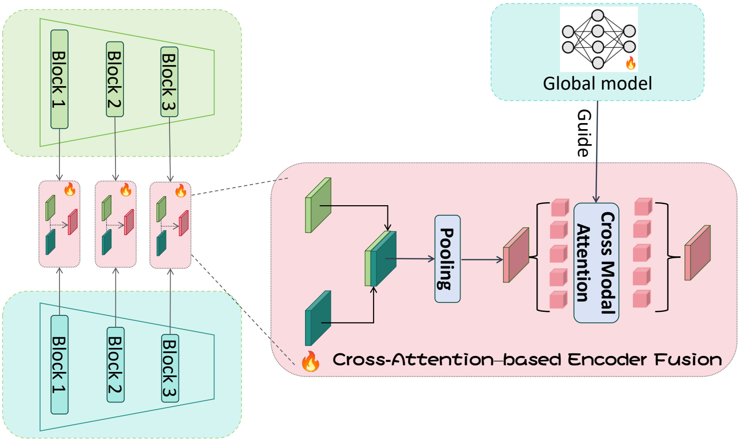

Fig. 2 Multimodal encoder fusion process

3 Methodology

As shown in Figure 1, we design a cross-attention-based multimodal fusion strategy to aggregate encoder representations across modalities. By combining modality-specific encoder parameters uploaded from clients with a public full-modality dataset, the server learns a multimodal decoder that integrates heterogeneous representations across modalities and strengthens the global model. Each client participant maintains a personalized model composed of an encoder and a decoder to generate client-specific predictions. This design allows each client to obtain outputs that better align with its own modality requirements. These schemes are detailed in the following sections.

3.1 Cross-Attention-Based Multimodal Fusion

To preserve privacy, clients upload only encoder parameters while keeping decoders local. During training, contrastive learning is used to align local and global representations, mitigating missing-modality issues and improving robustness.To integrate heterogeneous modalities, we adopt a cross-attention mechanism for multimodal fusion [7], enabling effective modeling of complementary modality information.

Cross-attention allows the model to reweight modality-specific features and produce fused representations that better reflect complementary modality cues.For clarity, we describe the fusion process using two modalities, denoted as α and β. After passing through their respective encoders, the resulting feature embeddings are represented as $\Phi(\alpha)$ and $\Phi(\beta)$.

A cross-modal attention mechanism is then employed to model the interactions between these representations and facilitate multimodal integration, as illustrated in Figure 2.

In the attention mechanism, we treat the features of the first modality α as the query (Query), and the features of the second modality β as the key (Key) and value (Value). The cross-modal attention weights are computed as follows:

$$\mathrm{Attention}\big(\Phi(\alpha), \Phi(\beta)\big) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{3-1}$$

$$\mathrm{Attention}\big(\Phi(\alpha), \Phi(\beta)\big) = \mathrm{softmax}\left(\frac{W_q\, \Phi(\alpha)\, \Phi(\beta)^\top\, W_k^\top}{\sqrt{d_k}}\right) W_v\, \Phi(\beta), \tag{3-2}$$

$$\mathrm{Attention}\big(\Phi(\alpha), \Phi(\beta)\big) \to A_{\mathrm{crossattn}}\, W_v\big(\Phi(\beta)\big) \to \hat{\Phi}(\beta). \tag{3-3}$$

where $Q = W_q\, \Phi(\alpha),\ K = W_k\, \Phi(\beta)$, and $V = W_v\, \Phi(\beta)$ denote the query, key, and value, respectively. Here, W_q, W_k, and W_v are learnable weight matrices, and d_k is the dimensionality of the key. The resulting attention weight matrix $A_{\mathrm{crossattn}}$ is used to reweight modality features and produce $\hat{\Phi}(\beta)$, enabling effective cross-modal information aggregation.

After multimodal attention weighting, we concatenate the reweighted feature of the second modality with the feature of the first modality to form the fused representation:

$$\Phi' = \big[\Phi(\alpha), \hat{\Phi}(\beta)\big]. \tag{3-4}$$

The fused representation is fed into the decoder to generate the fused image $\hat{I}$.

To evaluate fusion quality, we introduce a content loss and a structural similarity loss. The content loss measures pixel-level reconstruction error between the generated image and the target image:

$$\mathcal{L}_{\mathrm{recon}} = \left\|\hat{I} - I\right\|_2^2. \tag{3-5}$$

To further preserve structural information, we adopt the structural similarity index (SSIM), defined as

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)\,(2\sigma_{xy} + C_2)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)}, \tag{3-6}$$

based on which the SSIM loss is given by

$$\mathcal{L}_{\mathrm{SSIM}} = 1 - \mathrm{SSIM}(\hat{I}, I). \tag{3-7}$$

To encourage cross-modal feature alignment, we further introduce a cross-modality consistency loss:

$$\mathcal{L}_{\mathrm{consistency}} = \|\Phi(\alpha) - \Phi(\beta)\|_2^2\,. \tag{3-8}$$

Finally, to refine the fused representation and avoid excessive dominance of any single modality, we introduce a weighted fusion loss:

$$\mathcal{L}_{\text{fusion}} = \left\| \hat{\Phi}(\alpha, \beta) - \Phi' \right\|_2^2. \tag{3-9}$$

Together, these objectives facilitate effective information integration during cross-modal fusion while preserving visual quality and modality balance.

While the aforementioned objectives emphasize fusion quality at the representation level, the fused images are ultimately intended for downstream tasks such as segmentation and detection. As illustrated by the "Downstream model guide" in Figure 2, we further introduce a task-aware guidance strategy without directly performing end-to-end back-propagation from the downstream task.

Specifically, we reuse the feature extractor of the downstream task model as a task encoder $h(\cdot)$ and impose a consistency constraint between the fused image $\hat{I}$ and the full-modality reference image I in the task feature space:

$$\mathcal{L}_{\text{guide}} = \left\| h(\hat{I}) - h(I) \right\|_2^2. \tag{3-10}$$

This objective encourages the cross-modal attention module to preserve modality cues that are most relevant to downstream segmentation and detection.

3.2 Global-Local Contrastive Learning Strategy

In MFL, clients must accommodate heterogeneous modality inputs while preserving privacy and personalization. We use modality-level contrastive learning on the client side to reduce modality heterogeneity and aggregation drift. This strategy promotes alignment between local representations and the global model while maintaining client-specific characteristics.

In practical settings, clients often exhibit diverse data distributions, making it difficult for a single global model to capture personalized characteristics. We therefore incorporate modality contrastive learning into client-side training. By jointly leveraging local and global representations, this strategy allows the client encoder to retain local data traits while progressively aligning with the global model.

On each client, the encoder is optimized with a modality contrastive learning objective [5]. Let $i_{\text{local}}^{(k)}$ and $i_{\text{global}}^{(k)}$ denote the local and global image representations at communication round k, respectively, and let $i_{\text{prev}}^{(k)}$ represent the local representation from the previous round. The client encoder is trained by minimizing the following objective:

$$\mathcal{L}_{\text{intra}}^{(k)} = -\log\left(\frac{\exp\left(i_{\text{local}}^{(k)} \cdot i_{\text{global}}^{(k)}/\tau\right)}{\exp\left(i_{\text{local}}^{(k)} \cdot i_{\text{global}}^{(k)}/\tau\right) + \exp\left(i_{\text{local}}^{(k)} \cdot i_{\text{prev}}^{(k)}/\tau\right)}\right), \tag{3-11}$$

where τ is a temperature parameter. This objective encourages the local encoder to align its representations with the global model across communication rounds, thereby reducing modality-induced drift.

By contrast, the client-side decoder is trained exclusively on local data and remains entirely private. It produces task-specific outputs tailored to the client's data distribution, thereby supporting personalization without exposing sensitive information.

From the server viewpoint, modality contrastive learning promotes alignment of client-side representations in the feature space. With reduced cross-client discrepancies, the server-side cross-modal attention mechanism can more accurately model inter-modality dependencies, leading to stronger multimodal integration.

Modality contrastive learning strikes a balance between global consistency and client-level personalization. Sharing only encoder parameters while retaining local decoders ensures privacy and strengthens the robustness of the MFL framework.

3.3 Personalized MFL Training

By decoupling global representation learning and local task-specific optimization, clients can participate in collaborative training without exposing sensitive data while maintaining adaptability to heterogeneous multimodal inputs. The overall training procedure consists of the following stages.

(1) Initialization stage: At the beginning of each communication round, the server initializes a global model composed of a shared encoder and a multimodal fusion decoder and distributes the encoder parameters to all clients. Each client initializes its personalized decoder locally. During this stage, only encoder parameters are shared, while decoder parameters remain entirely local to ensure privacy preservation.

(2) Client training stage: Each client performs local training using its private data. The encoder is optimized with modality contrastive learning to encourage alignment between local and global representations, thereby reducing modality-induced drift. Meanwhile, the client-side decoder is trained in a personalized manner on local data to generate task-specific outputs.

(3) Global model update: After local training, clients upload only the updated encoder parameters. To train a shared multimodal fusion decoder, the server performs cross-attention–based multimodal fusion using the received encoders and a public full-modality dataset,. The server then aggregates the client encoders using federated averaging, and broadcasts the global encoder to all clients.

(4) Iterative optimization: The above steps are repeated over multiple communication rounds until convergence or a predefined number of rounds is reached. Through iterative client-side personalization and server-side multimodal fusion and aggregation, the proposed framework progressively improves global fusion capability while preserving client-specific adaptation under modality heterogeneity.

Table 1 mDSC (%) on BraTS 2020 for image segmentation: personalized client performance (by modality) and server performance.

Method	**T1c**	**T1**	**T2**	**FLAIR**	**Avg**	**Server**
FedAvg	55.38	29.52	47.06	47.14	44.80	60.73
FedNorm	61.25	32.56	49.13	50.07	48.28	64.93
FedMSplit	62.33	39.25	52.58	49.92	51.56	67.85
FedIoT	64.07	42.53	54.15	51.72	53.14	67.32
CreamFL	59.36	40.13	50.02	48.25	49.47	67.23
FedMCA	**65.03**	**46.38**	**54.56**	**53.82**	**54.94**	**69.38**

4 Experiment

4.1 Experiment Setting

Dataset: We evaluate our method on two tasks: object detection on M3FD [8] and medical image segmentation on BraTS 2020 [9]. The datasets contain paired infrared/visible images and multimodal MRI scans, respectively. All datasets are split into training/validation/test sets with a ratio of 8:1:1, and the training data are partitioned across the server and clients.

Network architecture: For object detection, we adopt a hierarchical encoder–decoder architecture for feature extraction and reconstruction, and employ YOLOv5 [10] as the detector.

For medical image segmentation, we adopt RFNet [11] as the backbone, where its modality-specific encoder and regularization decoder are used as the client-side encoder and decoder, respectively.

Experimental settings: We set the batch size to 64, learning rate to 1e-4, and communication rounds to 300, with 5 local epochs per round. We compare FedMCA with FedAvg [12], FedNorm [13], FedMSplit [14], FedIoT [15], and CreamFL [5]. All baselines follow their default settings.

4.2 Image Segmentation Task

We evaluate the proposed framework on BraTS 2020 and report modality-specific, average, and global performance (Table 1). We use mDSC (%) as the evaluation metric.As shown in Table 1, FedMCA achieves the best performance across all settings, with improvements in both modality-specific and global mDSC. Fig. 3 further demonstrates that FedMCA produces more complete and structurally consistent segmentation results.

Compared with FedMCA, all baselines achieve inferior performance. FedAvg struggles with multimodal fusion, while other methods show limited gains due to insufficient cross-modality interaction or sensitivity to modality imbalance.

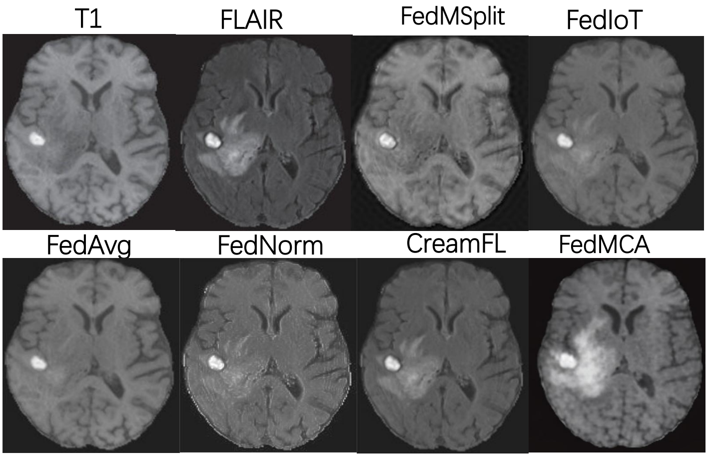

Fig. 3 Visual comparison of medical image fusion

Table 2 AP@0.5 (%) on M3FD for object detection under cross-modality images with the same annotations (server-side performance).

Method	Bus	Car	Lam	Mot	Peo	Tru	Avg
FedAvg	68.57	78.69	60.17	53.42	70.91	55.77	64.62
FedNorm	69.65	82.51	74.70	58.72	69.61	58.78	68.99
FedMSplit	68.15	81.94	74.95	**62.80**	69.41	59.04	69.38
FedIoT	68.92	81.79	**77.41**	59.34	69.41	59.98	69.48
CreamFL	61.33	**84.76**	77.13	59.34	71.52	58.65	70.45
FedMCA	**72.90**	82.88	76.98	62.62	**72.60**	**61.88**	**71.94**

4.3 Object Detection Task

We evaluate on the M3FD dataset using YOLOv5 and report AP@0.5 (%) (Table 2). FedMCA achieves the best performance, improving mean AP@0.5 by 7.32% over FedAvg, demonstrating effective multimodal fusion. Compared with FedMCA, all baselines show inferior performance due to limited cross-modal modeling or reliance on conventional aggregation strategies.FedMSplit suffers from increased complexity and reduced efficiency under heterogeneous settings, while FedIoT relies on conventional aggregation and lacks effective cross-modal fusion. CreamFL requires distributing server-side data to clients, which raises potential privacy concerns and increases the training burden on clients. In contrast, FedMCA consistently demonstrates stronger multimodal integration and achieves higher detection accuracy than both traditional FL approaches and recent MFL baselines.

4.4 Ablation Study

We conduct an ablation study to assess the individual contributions of the cross-attention fusion module, modality contrastive learning, and task-aware guidance in FedMCA. We evaluate four variants:

Table 3 Ablation study of FedMCA components: AP@0.5 (%) on M3FD and mDSC (%) on BraTS 2020

Dataset	FedAvg	FedMCA^1	FedMCA^2	FedMCA^3	FedMCA
M3FD	64.62	70.01	66.39	70.50	**71.94**
BraTS 2020	60.73	67.45	62.34	68.66	**69.38**

(i) **FedMCA**^{1}, which employs cross-attention for multimodal fusion but excludes modality contrastive learning and task-aware guidance;

(ii) **FedMCA**^{2}, which incorporates modality contrastive learning while replacing cross-attention with a standard fusion strategy;

(iii) **FedMCA**^{3}, which combines cross-attention fusion with task-aware guidance during training but omits modality contrastive learning;

(iv) **FedMCA**, the full model integrating all components.

As shown in Table 3, cross-attention fusion significantly improves performance, while contrastive learning and task-aware guidance provide additional gains. The full model (FedMCA) achieves the best results on both M3FD and BraTS 2020, demonstrating the effectiveness of all components.

5 Conclusion

In this paper, we propose a personalized MFL framework by combining global–local contrastive learning with cross-attention aggregation.In particular, the cross-modal feature alignment is proposed with adaptive fusion to reduce inconsistencies caused by heterogeneous and incomplete modalities, thus strengthening the stability and generalization of the global model. On the client side, modality contrastive learning aligns local and global representations to enhance adaptability without sharing sensitive information. In this way, it improves multimodal fusion while preserving client-specific modality characteristics. We evaluate the proposed framework under heterogeneous multimodal settings on two downstream tasks: medical image segmentation and object detection. Across all benchmarks, FedMCA consistently outperforms representative MFL baselines under different modality heterogeneity and missing-modality scenarios.

Despite its performance, we acknowledge that FedMCA, similar to some existing approaches, relies on the availability of a full-modality public dataset at the server for feature alignment. While this is a common prerequisite for cross-modal synchronization, it may pose challenges in scenarios where public auxiliary data is scarce. Future research will focus on relaxing this assumption by exploring data-free knowledge distillation or generative data augmentation to further enhance privacy and deployment flexibility.

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